Newton's second law: vary force

LEARNING TARGET · S4 I can measure how acceleration changes with force at constant mass — and F = ma emerges.

A · KEY VOCABULARY

Driving force	The hanging weight that pulls the cart system.
Total moving mass	Held constant: move masses from cart to hanger; never add new ones.
Proportional	Double the net force → double the acceleration.
Friction offset	Friction lowers the net force, so real data can sit below the ideal line.

B · KEY EQUATIONS + UNITS

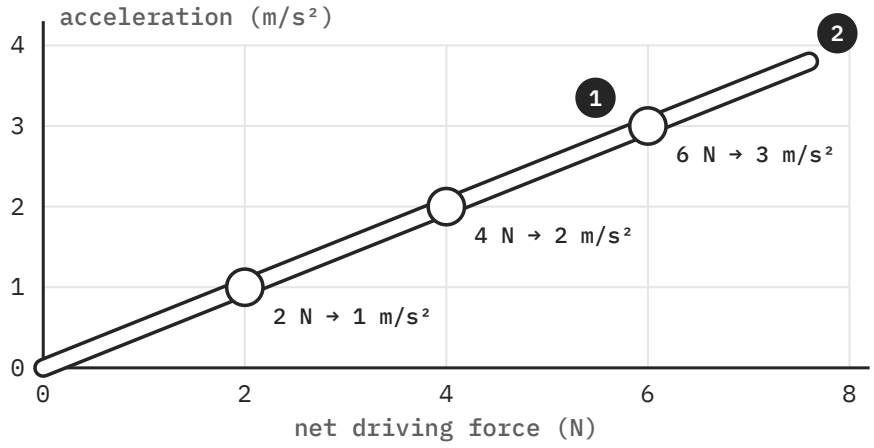
$$a \propto F \text{ and } a \propto 1/m$$
$$\rightarrow a = F \div m$$

slope of a vs. F = 1/m

force	newtons (N)
a	m/s²

C · COLOR IT · PLOT THE PATTERN

Ideal 2 kg system, illustrative.



☐ **1 · PENCIL** Data points

☐ **2 · GREEN** Trend line through zero

D · WORKED EXAMPLE

In an ideal 2 kg system, 2 N gives 1 m/s². Predict the acceleration at 4 N with the same mass.

GIVEN	EQUATION	WORK	ANSWER
F = 2 N → a = 1 m/s² new F = 4 N same mass	a ∝ F (same mass)	4 N ÷ 2 N = 2× the force a = 2 × 1 m/s²	**2 m/s²** acceleration doubles

E · THE 2026-XJ THREAD

DAY 12 OF 22 · NEW TOOL: NET FORCE

A bigger net force gives a bigger acceleration, in proportion. That's the rule analysts use to ask how hard XJ would have to be pushed to change course.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else